

The exponential family

- Exponential families
- Hypothesis tests

The exponential family of distributions

The **exponential family** of distributions contains the distributions whose probability functions attain the form

$$p(w|\theta) = \xi(w) \exp \left(\sum_{s=1}^S h_s(\theta) T_s(w) - A(\theta) \right) = \xi(w) \exp \left(\begin{pmatrix} h_1(\theta) \\ h_2(\theta) \\ \vdots \\ h_S(\theta) \end{pmatrix} \cdot \begin{pmatrix} T_1(w) \\ T_2(w) \\ \vdots \\ T_S(w) \end{pmatrix} - A(\theta) \right)$$

Here, $\xi(w)$ and $T_{1:S}(w)$ are **distribution-specific** functions of the variable; while, $h_{1:S}(\theta)$ and $A(\theta)$ are **distribution-specific** functions of the parameters, and S is the dimension of the parameter

Example: The Poisson distribution

$$w \sim \text{Poisson}(\mu), \quad p(w|\mu) = \frac{\mu^w}{w!} e^{-\mu} = \frac{1}{w!} \exp(w \log \mu - \mu)$$

belongs to the exponential family with

$$S = 1, \quad h(\mu) = \log \mu, \quad T(w) = w, \quad A(\mu) = \mu$$

The exponential family of Bayesian models

The **exponential family** of Bayesian models contains all models whose priors and likelihoods attain the form

$$p(\theta) = \zeta(\nu, \chi_{1:S}) \exp \left(\sum_{s=1}^S h_s(\theta) \chi_s - \nu A(\theta) \right)$$
$$p(w_n | \theta) = \xi(w_n) \exp \left(\sum_{s=1}^S h_s(\theta) T_s(w_n) - A(\theta) \right), \quad n = 1, \dots, N$$

Here, $\xi(w_n)$ and $T_{1:S}(w_n)$ are **model-specific** functions of the measurements; while, $h_{1:S}(\theta)$ and $A(\theta)$ are **model-specific** functions of the parameters, S is the dimension of the parameter, $\chi_{1:S}, \nu$ are the hyper-parameters, and $\zeta(\chi_{1:S}, \nu)$ is a **model-specific** function of the hyper-parameters

The exponential family of Bayesian models

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Example: The Gamma-Poisson model

$$\mu \sim \text{Gamma}(\phi, \psi), \quad p(\mu) = \frac{\mu^{\phi-1} e^{-\mu/\psi}}{\Gamma(\phi) \psi^\phi} = \frac{1}{\Gamma(\phi) \psi^\phi} \exp \left((\phi-1) \log \mu - \frac{\mu}{\psi} \right)$$

$$w | \mu \sim \text{Poisson}(\mu), \quad p(w | \mu) = \frac{\mu^w e^{-\mu}}{w!} = \frac{1}{w!} \exp(w \log \mu - \mu)$$

belongs to the exponential family with

$$S = 1, \quad h(\mu) = \log \mu, \quad T(w) = w, \quad A(\mu) = \mu, \quad \chi = \phi - 1, \quad \nu = 1/\psi$$

Bayesian updates within the exponential family

We consider a Bayesian model in the **exponential family**

$$p(\theta) = \zeta(\nu, \chi_{1:S}) \exp \left(\sum_{s=1}^S h_s(\theta) \chi_s - \nu A(\theta) \right)$$
$$p(w_n | \theta) = \xi(w_n) \exp \left(\sum_{s=1}^S h_s(\theta) T_s(w_n) - A(\theta) \right), \quad n = 1, \dots, N$$

Our posterior is given by

$$\begin{aligned} p(\theta | w_{1:N}) &\propto p(w_{1:N} | \theta) p(\theta) \\ &= \left[\prod_{n=1}^N p(w_n | \theta) \right] p(\theta) \\ &= \left[\prod_{n=1}^N \xi(w_n) \exp \left(\sum_{s=1}^S h_s(\theta) T_s(w_n) - A(\theta) \right) \right] \zeta(\nu, \chi_{1:S}) \exp \left(\sum_{s=1}^S h_s(\theta) \chi_s - \nu A(\theta) \right) \\ &\propto \left[\prod_{n=1}^N \exp \left(\sum_{s=1}^S h_s(\theta) T_s(w_n) - A(\theta) \right) \right] \exp \left(\sum_{s=1}^S h_s(\theta) \chi_s - \nu A(\theta) \right) \end{aligned}$$

Bayesian updates within the exponential family

We consider a Bayesian model in the **exponential family**

$$p(\theta) = \zeta(\nu, \chi_{1:S}) \exp \left(\sum_{s=1}^S h_s(\theta) \chi_s - \nu A(\theta) \right)$$
$$p(w_n | \theta) = \xi(w_n) \exp \left(\sum_{s=1}^S h_s(\theta) T_s(w_n) - A(\theta) \right), \quad n = 1, \dots, N$$

Our posterior is given by

$$\begin{aligned} p(\theta | w_{1:N}) &\propto \left[\prod_{n=1}^N \exp \left(\sum_{s=1}^S h_s(\theta) T_s(w_n) - A(\theta) \right) \right] \exp \left(\sum_{s=1}^S h_s(\theta) \chi_s - \nu A(\theta) \right) \\ &= \exp \left(\sum_{n=1}^N \left[\sum_{s=1}^S h_s(\theta) T_s(w_n) - A(\theta) \right] + \sum_{s=1}^S h_s(\theta) \chi_s - \nu A(\theta) \right) \\ &= \exp \left(\sum_{s=1}^S h_s(\theta) \left(\chi_s + \sum_{n=1}^N T_s(w_n) \right) - (\nu + N) A(\theta) \right) \\ &= \exp \left(\sum_{s=1}^S h_s(\theta) \chi'_s - \nu' A(\theta) \right) \end{aligned}$$

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Our posterior is given by

$$\begin{aligned} p(\theta | w_{1:N}) &\propto \exp \left(\sum_{s=1}^S h_s(\theta) \chi'_s - \nu' A(\theta) \right) \\ &= \frac{\zeta(\nu', \chi'_{1:S})}{\zeta(\nu', \chi'_{1:S})} \exp \left(\sum_{s=1}^S h_s(\theta) \chi'_s - \nu' A(\theta) \right) \\ &\propto \zeta(\nu', \chi'_{1:S}) \exp \left(\sum_{s=1}^S h_s(\theta) \chi'_s - \nu' A(\theta) \right) \end{aligned}$$

Bayesian updates within the exponential family

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Our posterior is given by

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Our model is **conjugate** and the posterior **hyper-parameters** are

$$\chi'_s = \chi_s + \sum_{n=1}^N T_s(w_n) \quad \nu' = \nu + N$$

This means that our prior is **equivalent** to a pseudo-dataset with ν pseudo-measurements

Hypothesis tests with uncertain parameters

We consider a hypothesis testing problem with **uncertain parameters**

$$\theta_{H_1} \sim P_{H_1}$$

$$\theta_{H_2} \sim P_{H_2}$$

$$\vdots$$

$$\theta_{H_M} \sim P_{H_M}$$

$$\eta \sim \text{Cat}_{H_1, H_2, \dots, H_M}(\pi_{H_1}, \pi_{H_2}, \dots, \pi_{H_M})$$

$$w|\eta, \theta_{H_1}, \theta_{H_2}, \dots, \theta_{H_M} \sim L_{\eta}^{\theta_h}$$

and seek a **decision** based on the **marginal posterior**

$$\eta_* = \operatorname{argmax}_{\eta} p(\eta|w)$$

In turn, this requires evaluation of the **marginal likelihoods**

$$\ell_{\eta} = \int d\theta_{\eta} L_{\eta}^{\theta_{\eta}}(w) P_{\eta}(\theta_{\eta})$$

Hypothesis tests within the exponential family

For each hypothesis, we consider **prior/likelihood pairs** within the exponential family

$$P(\theta) = \zeta(\nu, \chi_{1:S}) \exp \left(\sum_{s=1}^S h_s(\theta) \chi_s - \nu A(\theta) \right) \quad L^\theta(w) = \xi(w) \exp \left(\sum_{s=1}^S h_s(\theta) T_s(w) - A(\theta) \right)$$

Our **marginal likelihood** integrals are

$$\begin{aligned} \ell &= \int d\theta L^\theta(w) P(\theta) \\ &= \int d\theta \xi(w) \exp \left(\sum_{s=1}^S h_s(\theta) T_s(w) - A(\theta) \right) \zeta(\nu, \chi_{1:S}) \exp \left(\sum_{s=1}^S h_s(\theta) \chi_s - \nu A(\theta) \right) \\ &= \xi(w) \zeta(\nu, \chi_{1:S}) \int d\theta \exp \left(\sum_{s=1}^S h_s(\theta) (\chi_s + T_s(w)) - (\nu + 1) A(\theta) \right) \\ &= \xi(w) \zeta(\nu, \chi_{1:S}) \int d\theta \exp \left(\sum_{s=1}^S h_s(\theta) \chi'_s - \nu' A(\theta) \right) \end{aligned}$$

Hypothesis tests within the exponential family

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Our **marginal likelihood** integrals are

$$\begin{aligned} \ell &= \xi(w) \zeta(\nu, \chi_{1:S}) \int d\theta \exp \left(\sum_{s=1}^S h_s(\theta) \chi'_s - \nu' A(\theta) \right) \\ &= \xi(w) \zeta(\nu, \chi_{1:S}) \int d\theta \frac{\zeta(\nu', \chi'_{1:S})}{\zeta(\nu', \chi'_{1:S})} \exp \left(\sum_{s=1}^S h_s(\theta) \chi'_s - \nu' A(\theta) \right) \\ &= \xi(w) \frac{\zeta(\nu, \chi_{1:S})}{\zeta(\nu', \chi'_{1:S})} \int d\theta \zeta(\nu', \chi'_{1:S}) \exp \left(\sum_{s=1}^S h_s(\theta) \chi'_s - \nu' A(\theta) \right) \\ &= \xi(w) \frac{\zeta(\nu, \chi_{1:S})}{\zeta(\nu', \chi'_{1:S})} \end{aligned}$$

Signal detection

We consider the **detection** of a signal with certain intensity levels $\mu_{1:M}$ transmitted over a noisy channel with uncertain intensity-dependent noise

Signal detection becomes a **hypothesis testing** problem

$$\begin{aligned}\eta &\sim \text{Cat}_{1:M} \left(\frac{1}{M}, \frac{1}{M}, \dots, \frac{1}{M} \right) \\ \tau_m &\sim \text{Gamma}(\phi, \psi), & m = 1, \dots, M \\ w_n | \eta, \tau_{1:M} &\sim \text{Normal} \left(\mu_\eta, \frac{1}{\tau_\eta} \right), & n = 1, \dots, N\end{aligned}$$

Our **marginal posterior** is given by

$$\begin{aligned}p(\eta | w_{1:N}) &= \int_0^\infty d\tau_1 \int_0^\infty d\tau_2 \cdots \int_0^\infty d\tau_M p(\eta, \tau_{1:M} | w_{1:N}) \\ &\propto \int_0^\infty d\tau_1 \int_0^\infty d\tau_2 \cdots \int_0^\infty d\tau_M p(w_{1:N} | \eta, \tau_{1:M}) p(\eta, \tau_{1:M}) \\ &= \int_0^\infty d\tau_1 \int_0^\infty d\tau_2 \cdots \int_0^\infty d\tau_M \left[\prod_{n=1}^N p(w_n | \eta, \tau_{1:M}) \right] p(\eta) \left[\prod_{m=1}^M p(\tau_m) \right]\end{aligned}$$

Signal detection

Our **marginal posterior** is given by

$$\begin{aligned} p(\eta|w_{1:N}) &\propto \int_0^\infty d\tau_1 \int_0^\infty d\tau_2 \cdots \int_0^\infty d\tau_M \left[\prod_{n=1}^N p(w_n|\eta, \tau_{1:M}) \right] p(\eta) \left[\prod_{m=1}^M p(\tau_m) \right] \\ &= \int_0^\infty d\tau_1 \int_0^\infty d\tau_2 \cdots \int_0^\infty d\tau_M \left[\prod_{n=1}^N p(w_n|\eta, \tau_\eta) \right] p(\eta) \left[\prod_{m=1}^M p(\tau_m) \right] \\ &= p(\eta) \left(\int_0^\infty d\tau_\eta \left[\prod_{n=1}^N p(w_n|\eta, \tau_\eta) \right] p(\tau_\eta) \right) \underbrace{\int_0^\infty d\tau_1 \int_0^\infty d\tau_2 \cdots \int_0^\infty d\tau_M \prod_{m \neq \eta} p(\tau_m)}_{m \neq \eta} \\ &= p(\eta) \int_0^\infty d\tau_\eta \left[\prod_{n=1}^N p(w_n|\eta, \tau_\eta) \right] p(\tau_\eta) \\ &= p(\eta) \int_0^\infty d\tau_\eta \left[\prod_{n=1}^N \text{Normal} \left(w_n; \mu_\eta, \frac{1}{\tau_\eta} \right) \right] \text{Gamma}(\tau_\eta; \phi, \psi) \end{aligned}$$

Signal detection

Our **marginal posterior** is given by

$$\begin{aligned}
 p(\eta|w_{1:N}) &\propto p(\eta) \int_0^\infty d\tau_\eta \left[\prod_{n=1}^N \text{Normal} \left(w_n; \mu_\eta, \frac{1}{\tau_\eta} \right) \right] \text{Gamma}(\tau_\eta; \phi, \psi) \\
 &= p(\eta) \int_0^\infty d\tau_\eta \left[\prod_{n=1}^N \sqrt{\frac{\tau_\eta}{2\pi}} e^{-\tau_\eta \frac{(w_n - \mu_\eta)^2}{2}} \right] \frac{1}{\Gamma(\phi)\psi^\phi} \tau_\eta^{\phi-1} e^{-\tau_\eta/\psi} \\
 &= p(\eta) \int_0^\infty d\tau_\eta \left(\frac{\tau_\eta}{2\pi} \right)^{N/2} e^{-\tau_\eta \frac{\sum_{n=1}^N (w_n - \mu_\eta)^2}{2}} \frac{1}{\Gamma(\phi)\psi^\phi} \tau_\eta^{\phi-1} e^{-\tau_\eta/\psi} \\
 &= \frac{p(\eta)}{(2\pi)^{N/2} \Gamma(\phi)\psi^\phi} \int_0^\infty d\tau_\eta \tau_\eta^{\phi + \frac{N}{2} - 1} e^{-\tau_\eta \left(\frac{1}{\psi} + \frac{\sum_{n=1}^N (w_n - \mu_\eta)^2}{2} \right)} \\
 &= \frac{p(\eta)}{(2\pi)^{N/2} \Gamma(\phi)\psi^\phi} \int_0^\infty d\tau_\eta \tau_\eta^{\phi' - 1} e^{-\frac{\tau_\eta}{\psi'_\eta}} \\
 &= \frac{p(\eta)}{(2\pi)^{N/2} \Gamma(\phi)\psi^\phi} \int_0^\infty d\tau_\eta \frac{\Gamma(\phi')(\psi'_\eta)^{\phi'}}{\Gamma(\phi')(\psi'_\eta)^{\phi'}} \tau_\eta^{\phi' - 1} e^{-\frac{\tau_\eta}{\psi'_\eta}} \\
 &= \frac{p(\eta)}{(2\pi)^{N/2}} \frac{\Gamma(\phi')(\psi'_\eta)^{\phi'}}{\Gamma(\phi)\psi^\phi}
 \end{aligned}$$

Signal detection

Our **marginal posterior** is given by

$$\begin{aligned} p(\eta|w_{1:N}) &\propto \frac{p(\eta)}{(2\pi)^{N/2}} \frac{\Gamma(\phi')(\psi'_\eta)^{\phi'}}{\Gamma(\phi)\psi^\phi} \\ &= \frac{1/M}{(2\pi)^{N/2}} \frac{\Gamma(\phi')(\psi'_\eta)^{\phi'}}{\Gamma(\phi)\psi^\phi} \\ &\propto (\psi'_\eta)^{\phi'} \\ &= \left(\frac{1}{\psi} + \frac{\sum_{n=1}^N (w_n - \mu_\eta)^2}{2} \right)^{-\phi - N/2} \end{aligned}$$

Detection is determined by

$$\begin{aligned} \eta_* &= \operatorname{argmax}_\eta p(\eta|w_{1:N}) \\ &= \operatorname{argmax}_\eta \left(\frac{1}{\psi} + \frac{\sum_{n=1}^N (w_n - \mu_\eta)^2}{2} \right)^{-\phi - N/2} \end{aligned}$$